



Indian Institute of Information Technology, Surat

भारतीय सूचना प्रौद्योगिकी संस्थान, सूरत

(Institute of National Importance under Act of Parliament)

Department of Computer Science and Engineering

B.Tech- 4th Semester , Mid Examination

Subject & Code: Software Engineering (CS-401)

Date: 3-03-2025

Time : 1.30 pm - 3.00 p m

Marks: 20/30

Note: All questions are compulsory.

	SECTION A	Marks
1.	Assuming identical code base size and development team, rank the following software projects by their expected development duration. Provide a brief justification for your ranking. (a) A personal budgeting tool (b) A multiplayer online game (c) A hospital patient management system	3
2.	List five desirable characteristics of a good software requirements specification (SRS) document. Who are the different categories of users of the SRS document? In what ways is the SRS document useful to them?	5
3.	What do you understand about inconsistencies, anomalies, and incompleteness in an SRS document? Identify the inconsistencies, anomalies, and incompleteness in the following requirements of an academic activity automation software of an educational institute: "The semester performance of each student is computed as the average academic performance for the semester. The guardians of all students having poor performance record in the semester are mailed a letter informing about the poor performance of the ward and intimating that repetition of poor performance in the subsequent semester can lead to expulsion. The extracurricular activities of a student are also graded and taken into consideration for determination of the semester performance."	3
4.	Suppose you have estimated the nominal development time of a moderate-sized software product to be 5 months. You have also estimated that it will cost Rs. 50,000 to develop the software product. Now, the customer comes and tells you that he wants you to accelerate the delivery time by 10 per cent. How much additional cost would you charge the customer for this accelerated delivery? Irrespective of whether you take less time or more time to develop the product, you are essentially developing the same product. Why then does the effort depend on the duration over which you develop the product?	4

5.

Suppose you are a project manager of a software project requiring the following activities,

Activity	Immediate Follower	Estimated time in weeks
A	E,B	3
B	C,F	2
C	D,G	8
D	F	4
E	F	5
F	-	10
G	-	10

Draw the activity network representation of the tasks.
Identifies the sequence of activities that is on the critical path.
Identify the path that has the greatest slack time.

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SECTION B

6.

- a) List the goals of software engineering?
- b) Explain different software myths with respect to management, customer and practitioner with neat case studies.
- c) Why is the SDLC life cycle important for software engineers?
- d) What are the 4 types of maintenance in software engineering?
- e) How the waterfall and the prototyping models can be accommodated in the spiral process model. Analyze.

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